Java Live Coding Interview - Quick Revision

1. Prime Numbers Within a Range

• Loop from start to end.

• Skip numbers less than 2.

• Check divisibility from 2 to sqrt(n).

• If divisible → Not Prime.

• Else → Prime.

```
for(int i = start; i <= end; i++) {

    if(i < 2) continue;

    boolean isPrime = true;

    for(int j = 2; j <= Math.sqrt(i); j++) {

        if(i % j == 0) {
            isPrime = false;
            break;
        }
    }

    if(isPrime) {
        System.out.println(i + " Prime");
    }
}
```

2. Factorial

• Initialize factorial with 1.

• Multiply numbers from 1 to n.

```
int fact = 1;

for(int i = 1; i <= num; i++) {
    fact = fact * i;
}

System.out.println(fact);
```

3. GCD (Euclidean Algorithm)

• Use modulus operator for faster calculation.

• Repeat until b becomes 0.

```
while(b != 0) {

    int temp = b;
    b = a % b;
    a = temp;
}

System.out.println("GCD = " + a);
```

4. LCM

• Use formula: (a * b) / GCD

```
int lcm = (a * b) / gcd;

System.out.println(lcm);
```

5. Palindrome Number

• Reverse the number.

• Compare original and reversed number.

```
int temp = num;
int rev = 0;

while(num != 0) {

    int digit = num % 10;
    rev = rev * 10 + digit;
    num = num / 10;
}

if(temp == rev) {
    System.out.println("Palindrome");
}
```

6. Leap Year

• Divisible by 4 and not by 100.

• Or divisible by 400.

```
if((year % 4 == 0 && year % 100 != 0)
    || year % 400 == 0) {

    System.out.println("Leap Year");
}
```

7. Factors of a Number

• Check divisibility from 1 to n.

```
for(int i = 1; i <= num; i++) {

    if(num % i == 0) {
        System.out.print(i + " ");
    }
}
```

8. Fibonacci Series

• Store previous two numbers.

• Generate next using sum.

```
int a = 0, b = 1;

for(int i = 1; i <= n; i++) {

    System.out.print(a + " ");

    int next = a + b;
    a = b;
    b = next;
}
```

9. Reverse a Number

• Extract digit using % 10.

• Build reverse using rev * 10 + digit.

```
int rev = 0;

while(num != 0) {

    int digit = num % 10;

    rev = rev * 10 + digit;

    num = num / 10;
}

System.out.println(rev);
```

10. Second Largest Number in Array

• Track max and second max.

• Update both carefully.

```
int max = arr[0];
int secmax = arr[0];

for(int i = 1; i < arr.length; i++) {

    if(arr[i] > max) {

        secmax = max;
        max = arr[i];
    }

    else if(arr[i] > secmax && arr[i] != max) {

        secmax = arr[i];
    }
}

System.out.println(secmax);
```

Interview Tips

• Explain your logic while coding.

• Use meaningful variable names.

• Handle edge cases.

• Practice Scanner input.

• Know loops, arrays, methods, and conditions well.

• Stay calm even if you get stuck.